PROJECT REPORT FORMAT • TECHNICAL SUBMISSION

EV Insights: Electric Vehicle Market Performance & Analysis

Author / Assigned To:	Pranjal Joshi	**Team ID:**	Not Assigned
Project Scope:	Global EV Market & Specs	**Toolstack:**	Tableau Public, Python, React, SQL
Verified Dataset:	Multi-Region EV Models, Pricing & Battery Logs	**UI Standard:**	1920x1080 (Modern Responsive UI)

1. INTRODUCTION

1.1 Project Overview

The **EV INSIGHTS** project is an interactive data analytics and visualization web application designed to evaluate global electric vehicle (EV) market performance, manufacturer competition, pricing structures, battery technologies, and consumer satisfaction. As the adoption of electric vehicles accelerates worldwide, understanding complex automotive datasets becomes crucial for industry stakeholders, consumers, and analysts. This project leverages modern web technologies combined with integrated Tableau Public dashboards to transform raw, unstructured EV data into clear, interactive, and actionable visual insights.

1.2 Purpose

The purpose of this project is to model, clean, standardize, and visualize disparate automotive data into an ergonomic executive dashboard and narrative story. Key goals include establishing standardized performance metrics, dynamic manufacturer leaderboard filtering, quantifying battery capacity to driving range efficiency, and delivering an end-to-end analytical framework for sustainable transportation research.

2. IDEATION PHASE

2.1 Problem Statement

Framework Prompt	Customer Problem Statement Definition
I am	An Automotive Market Analyst, Clean-Tech Researcher, and Electric Vehicle Procurement Strategist.
I'm trying to	Evaluate global EV model specifications, analyze manufacturer pricing tiers, investigate battery capacity vs. driving range correlations, and compare brand market penetration.
but	Global EV telemetry and specification logs are fragmented across unstructured CSV files, pricing standards vary widely, and static spreadsheets lack cross-filtering and dynamic rank adjustments.
because	There is no centralized, interactive visual intelligence layer linking manufacturer pricing, battery technologies, and regional availability.
which makes me feel	Constrained by manual multi-sheet data reconciliation and hesitant during time-critical market strategy briefings.

2.2 Empathy Map Canvas

THINKS & FEELS	SAYS & DOES
Thinks: "Battery capacity and charging efficiency dictate consumer adoption just as heavily as upfront vehicle pricing." **Feels:** Overwhelmed by disparate specification logs across multiple brands; restricted by static tabular reports.	**Says:** "We need immediate visual distinction between premium luxury EVs and affordable commuter models across global regions." **Does:** Spends hours manually compiling specification sheets and cross-referencing price-to-range ratios.
PAINS: Data query latency, ambiguous specification labels, inability to dynamically focus between market leaders and emerging manufacturers.	**GAINS:** Sub-second diagnostic filtering, dynamic Top N manufacturer leaderboards, and clear narrative story slides for stakeholder presentations.

2.3 Brainstorming

S.No	Brainstormed Idea	Category	Feasibility	Status
1	Executive Dark Dashboard (1920x1080) with KPI Cards	Visual Analytics	High / High	Implemented
2	Dynamic Top N Manufacturer Parameter Slider	Interactive Controls	High / High	Implemented
3	Standardized Data Cleaning & Normalization Pipeline	Data Engineering	High / High	Implemented
4	Battery Capacity vs. Driving Range Efficiency Plot	Strategy Analytics	High / Medium	Implemented
5	Real-Time Charging Station Density API Stream	IoT / Live Data	Low / High	Future Scope

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer workflow advances across five stages: **1. Data Ingestion** (connecting multi-table EV specifications), **2. Macro Overview** (reviewing aggregate KPIs across manufacturers and pricing tiers), **3. Diagnostic Filtering** (isolating specific battery types and range brackets), **4. Strategy Benchmarking** (comparing cost-to-performance ratios), and **5. Executive Reporting** (presenting narrative story points to stakeholders).

3.2 Solution Requirement

- **Functional Requirements:** Dynamic date/model multi-select filters, standardized specification classification dropdowns, dynamic Top N parameter control, and cross-filtering between geographic market distribution and battery efficiency charts.
- **Non-Functional Requirements:** Fixed 1920x1080 Full HD responsive canvas, professional modern color palette with slate container borders, and sub-second query response via Tableau Hyper Data Extracts.

3.3 Data Flow Diagram

`Ingestion: Relational CSV Extracts (models, prices, battery capacity, range, manufacturers)` → `Cleaning Pipeline: Missing value imputation, string casting, duplicate removal` → `Tableau Hyper Extract: Pre-aggregated Fixed LOD metrics and parameters` → `Presentation Layer: 1920x1080 Executive Dashboard & Interactive Story.`

3.4 Technology Stack

Tableau Desktop & Tableau Public, Python (Pandas/NumPy), SQL database schema architecture, GitHub repository management, React/HTML5 web embed wrappers, and deployment tooling.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The design resolves data fragmentation through a structured 3-tier layout: Tier 1 houses the project title, executive KPI cards, and global filters; Tier 2 pairs the geographic market adoption map with the manufacturer reliability bar chart; Tier 3 organizes the dynamic leaderboard, pricing distributions, and battery capacity scatter plots.

4.2 Proposed Solution & 4.3 Solution Architecture

The solution implements an integrated dashboard adhering to a strict container grid. Global filters (Manufacturer, Price Range, and Top N Parameter) are placed in the upper control quadrant, broadcasting filter events across all worksheets simultaneously to ensure seamless interactive exploration.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project was executed across six structured sprints: **Week 1** (Problem Definition & Empathy Mapping), **Week 2** (Data Cleaning & Attribute Normalization), **Week 3** (Worksheet Development & Styling), **Week 4** (Fixed Canvas Assembly & Palette Calibration), **Week 5** (Tableau Story Narrative & Performance Testing), and **Week 6** (Tableau Public Deployment, GitHub Integration, and Video Walkthrough).

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Query latency profiling via the Tableau Performance Recorder showed a reduction in query execution times from 3,400 ms (live multi-table CSV queries) down to 380 ms utilizing a materialized Tableau Hyper Extract. Redundant nested containers were eliminated, achieving seamless sub-second filter updates across all connected charts.

7. RESULTS

7.1 Output Screenshots & Visualization Highlights

- **Verified Output Architecture:** Fixed Full HD dark canvas displaying all validated operational figures.
- **Executive KPI Cards:** Confirms total models analyzed, average market price, mean driving range, and active manufacturers.
- **Global Market Distribution Map:** Interactive geographic plot showcasing EV adoption density across key international markets.
- **Battery vs. Range Scatter Plot:** Quantifies efficiency trends, revealing linear scalability between kilowatt-hour capacity and total driving range.
- **Manufacturer Performance Leaderboard:** Dynamic Top N rankings displaying leading brands by volume, average pricing, and consumer ratings.
- **Pricing Distribution Histogram:** Evaluates affordability tiers across budget, mid-range, and luxury vehicle segments.

8. ADVANTAGES & DISADVANTAGES

Advantages: Single-pane visual diagnostics, sub-second query response, dynamic Top N customization, and guided executive storytelling.
Disadvantages: Fixed canvas dimensions require browser zoom adjustments on lower-resolution mobile screens; historical datasets lack real-time charging infrastructure telemetry.

9. CONCLUSION & 10. FUTURE SCOPE

Conclusion: The project successfully translates disparate global EV telemetry into an interactive, high-performance visual intelligence dashboard and narrative story in Tableau, providing actionable insights into vehicle pricing, battery efficiency, and manufacturer standing.
Future Scope: Integration of real-time charging station availability APIs, predictive machine learning models for future battery cost curves, and responsive mobile-optimized layouts.

11. APPENDIX

Source Code (Tableau Calculated Fields & Parameters)

```
// Dynamic Top N Manufacturer Ranking Filter
RANK(SUM([Total Models / Sales]), 'desc') <= [Top N Parameter]

// Standardized Price Tier Categorization
IF [Price (USD)] < 35000 THEN "Budget / Entry (< $35k)"
ELSEIF [Price (USD)] >= 35000 AND [Price (USD)] < 60000 THEN "Mid-Range ($35k - $60k)"
ELSEIF [Price (USD)] >= 60000 AND [Price (USD)] < 90000 THEN "Premium ($60k - $90k)"
ELSE "Luxury / High-End (> $90k)"
END

// Key Executive LOD Metrics
[Total Models Analyzed]: { FIXED [Manufacturer] : COUNTD([Model ID]) }
[Average Market Price]: { FIXED : AVG([Price (USD)]) }
[Mean Driving Range (km)]: { FIXED : AVG([Range (km)]) }
```

Dataset & Project Links

- **Dataset Source:** Global Electric Vehicle Market & Specification Repository (Cleaned & Standardized Records).
- **GitHub Repository:** github.com/joshipranjal2309-pj/EV_ANALYSIS
- **Live Tableau Public Demo:** Tableau Public Production Dashboard
- **Demonstration Walkthrough:** Technical screen recording submitted for project evaluation.